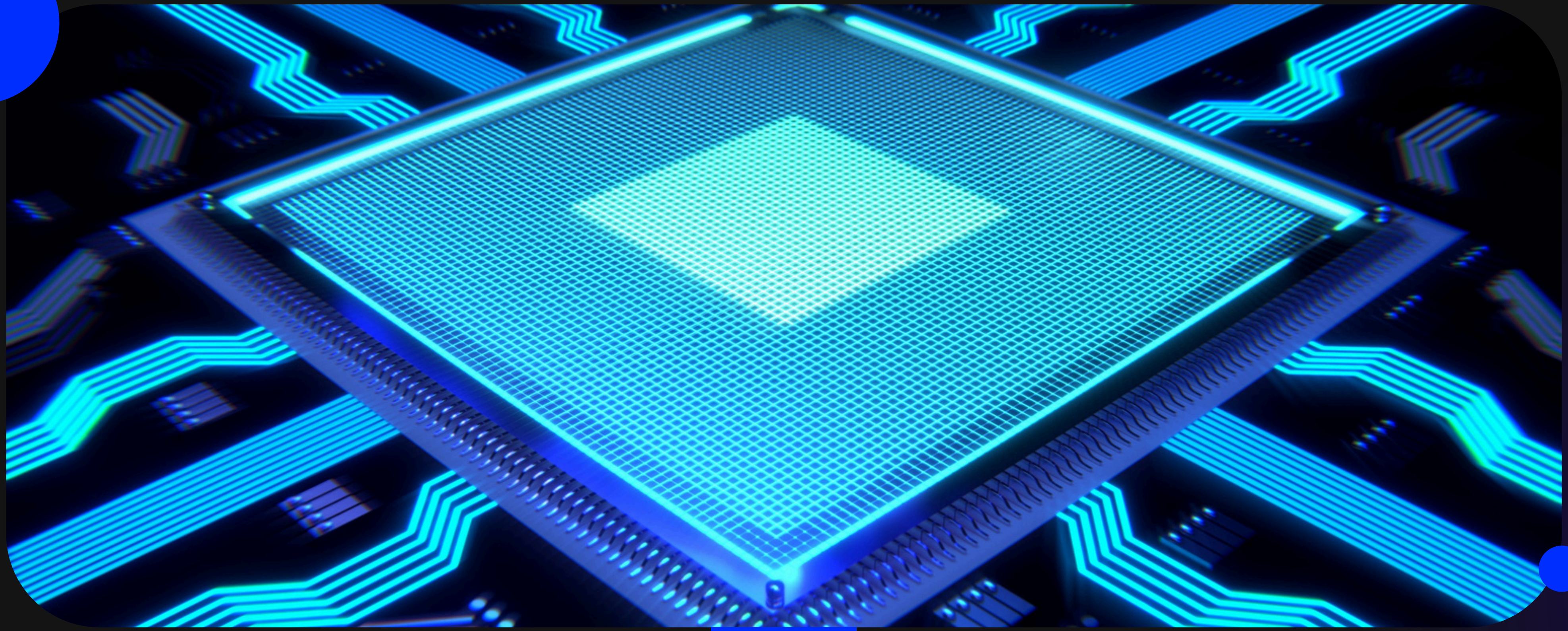
MOSHARAFA QUANTUM HACKATHON · TRACK 1

Quantum Simulation



Grand Finale Presentation

8 September 2026

In the spirit of Ali Moustafa Mosharafa — Egypt's pioneer of quantum theory, once called "a genius" by Einstein

THE BIG PICTURE

From Ethylene to Photosynthesis

One Flip of an Electron Powers Life & Light

- Photo-induced electronic excitations drive vision, photosynthesis, and next-gen solar cells.
- Simulating excited states classically explodes exponentially as molecules grow.
- Quantum Algorithms: VQE for ground states, QSE for excited states. Candidates for quantum advantage.
- Ethylene (C_2H_4) is the simplest molecule with a clean $\pi \rightarrow \pi^*$ excitation making it the perfect testbed.

The Energy Flow

Same cascade, three scales: retina · chloroplast · photovoltaic cell.

Track goal: build & benchmark a VQE+QSE pipeline on ethylene, then chart a path to a photosynthesis-scale system.

Our Pipeline: Molecule → Qubits → VQE → QSE

1. Build

Ethylene Hamiltonian, STO-3G, active space (2e,2o) at 0° & 90°

2. Map

Jordan-Wigner (4 qubits) & parity mapping with tapering (2 qubits)

3. VQE

UCCSD & hardware-efficient ansatz find the ground state

4. QEOM

We implemented QEOM instead of QSE for reasons explained in the report.

Every step cross-checked against a classical reference before trusting the quantum result.

Ethylene: Planar vs. 90° Twisted

- STO-3G basis, active space = 2 electrons in 2 spatial orbitals expandable to 2 electrons 4 orbitals.
- Two geometries: planar equilibrium and the 90° twisted structure along the C=C bond.
- Jordan-Wigner mapping → 4 qubits, 15–27 Pauli terms.
- Parity mapping + tapering → same physics on just 2 qubits.

4 → 2

Qubits, JW → parity

0.0000 Ha

Qiskit vs. PySCF ground state gap

Ansatz Choice

Method	Energy (Ha)	Qubits	Depth (parameters)
UCCSD (VQE)	−77.116628	4	112 (3)
Hardware-efficient (VQE)[1]	−77.116184	4	12 (24)
CASCI(2,2) [classical]	−77.116608	—	—

Note: the UCCSD ansatz reproduces the exact ground state energy essentially exactly, in just 3 parameters. The hardware efficient ansatz gets stuck near a local minimum, doubling its layers barely moves the needle. Physics informed structure beats brute force.

$\pi \rightarrow \pi^*$ Excitation Energy

Energy ordering and magnitude are consistent with a spin-resolved
CASCI(2,2) calculation performed as a cross-check (triplet at 4.91~eV,
singlet V-state at 14.24~eV, confirmed via
expectation values), allowing unambiguous spin assignment of the QEOM roots

The singlet V-state
(14.1~eV) substantially overestimates the experimental value of 7.6~eV; this
is a known limitation of the minimal STO-3G basis combined with a (2,2)
active space for this particular ionic transition, and is not attributable
to the quantum algorithm.

Every Number, Triple-Checked

CASCI (classical)

PySCF CASCI(2e,2o) singlet FCI — our ground truth.

Exact qubit diagonalization

Physical-sector filtered eigensolver on the mapped qubit Hamiltonian.

QEOM (alt. quantum method)

Equation-of-motion excited states from the same UCCSD VQE state.

All three agree to better than 10^{-6} eV. And the 2-qubit parity-tapered model reproduces the full 4-qubit energy exactly.

Noise-Aware Benchmarking Using Real Quantum Hardware

- We built the hardware-ready ingredient: a 2-qubit, tapered parity model that exactly reproduces the 4-qubit energy.
- We ran this circuit on real backend "ibm_marrakesh".
- We applied error mitigation techniques.
- We show how noise degrades the ground-state energy and the π→π* gap — and how much mitigation recovers.

WHAT WE HANDED IN

Three Files, One Story

team12.ipynb

The full notebook: theory walkthrough alongside the live code. Built to teach the method while it runs.

team12_classical_simulation.ipynb

The ground truth, walkthrough of the classical simulation and results.

team12_report.pdf

walking through the methodology behind every step, and the results.

Same pipeline, three views: learn it, run it, read about it.

Thank you!

Dedicated to Ali Moustafa Mosharafa – the man who dreamt in quantum theory decades before quantum computers.

Team 12 "Mosharafa" Members:

Rabee Banisalman

Ishak Tegguiche

Mariam Mohamed

Noor ul Ain

Jibril Abdullahi

Abibou Ndam